

AI in Museums

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9 July 2026

DISCLAIMER: I did not use AI in any way or form in the production of this piece. All errors and mistakes are completely my own and I am proud of these – they define me and are MY mistakes.

I have been asked to write an essay on AI in Museums - so let me begin at the beginning...

I used to have a little poster above my desk which read “*There are those who make things happen; those who watch things happen; and those who wonder what happened*”. I am here today to report that we are all in this last category as the world of AI has taken us by surprise with both the speed at which this has advanced and that none of us saw this coming.

Well, this is not quite true. I entered the AI world in 1974 while studying for my B.Sc. in Computer Science and then completed an M.Sc. in Computer Science with a focus on AI from 1977-78. In 1981 I started lecturing in Computer Science focusing on AI as one of my teaching responsibilities, and started an AI-based doctorate, which was too adventurous and which I eventually gave up, with the challenge being the lack of computing power to carry out the computations which were needed. I predicted this would only happen in around 50 years. And guess what! Those 50 years have now passed and here we are, and it is now my time to reflect on what we can do with this new technology, while knowing that we cannot predict the next months and years on what will happen next.

I had not given up on my love of AI, until now. I am rather disappointed about many aspects of the current AI revolution, mostly that it is great at simulating human outputs, but fails to truly understand what it is doing, and often this is based on its inability to explain reasoning, since the responses are aggregations of billions of other works, and perhaps everything written and published by us all is somewhere in those responses.

But do not get me wrong. I am truly impressed by AI, and I use it a lot every day, and this leads to my first observation, that AI is a tool, and for what it does is relatively cheap – and

for all of us to get ahead we must use this tool as much as possible, since everyone else is already doing this, and it is likely that they are already ahead of you.

So let me explore a second observation – that there are problems highly suited for AI, and others which are not suited, and it is with these where AI starts to make up things which have not happened – ever – and yet are used as though they are real.

Perhaps the most embarrassing of these hallucinations was with the recently published South African National Policy on AI, for which the contractor clearly used AI to generate it (and why not!), but a quick analysis showed that many of the citations used as the basis for the evidence pointed to non-existent research journals and articles. It was quickly withdrawn.

So, my third observation is then – what should we do and what should AI do – is it fair to ask an AI to generate a complete report for us with almost no input, compared to the previous approach in which we wrote the whole report ourselves, or is there some middle ground in which we can work together.

My focus in this article is not AI per se, but AI as applied to specific situations, and it truly has access to an exceptional amount of prior work and can make deductions through advanced pattern matching which are far beyond human capacity. The specific situation at hand is Museums, and specifically museum collections, and more generally heritage collections of any form.

So let me introduce how I think AI can be used in your museum or archives or library, and within the entire museum and heritage sector. I present five specific tasks which I believe are both within the power of the AIs, and which all fall beyond human capacity.

Knowing every Object

Every museum I have worked with has a host of items which are believed to be significant, and which were accessioned by previous curators, and there should have been a good reason to bring into the permanent collections. In the past there may not have been the more formalised accession policies that we expect of today's museums, and objects were just deposited on the doorstep and brought in without much thought. But it is expected that much of what is in the museum tells a story which is relevant to the historical record.

But the challenge is that now no-one knows where these items fit, and often even what they are!

AI can assist here, so long as the inventories of all museums are made available in some shared aggregator (more on this later) - with whatever cataloguing information and provenance information is available. AI services can explore this universal collection and find linkages which humans cannot do, either alone or with many others, and humans will never be able to do this. The AI can match up items from disparate places and can use this to discover what are the unknown items. And for AI this task is relatively simple, even without too much instruction and with incomplete information.

This specific problem has been one of my ideas for an AI-directed project, to find a way to discover the value hidden in those many objects which are in the museum but are unknown. I recall a talk by Nick Poole, the former CEO of Collections Trust, that in one museum he has worked with, more than 50% of the collection was marked as Miscellaneous, which is, in effect, saying that we know nothing about these items, and hide them under a generic label to allow us to forget about them without guilt.

Whereas incomplete inventories are a problem in almost all museums and are often targeted under “Banish the Backlog” initiatives, this problem arises when the items actually in the inventory are essentially unknown, so that no attempt at addressing a backlog can reach any outcome.

What a wonderful problem this is to give to our new AI friends...

Connecting Museums

I entered the Museum and Heritage sector through the back door. I was not trained in any of the heritage disciplines, but merely a computer scientist who likes difficult problems, and with a passion (as I already mentioned) for AI.

In 2005, I responded to a request for proposals from SAHRA (South African Heritage Resources Agency) for the development of a National Inventory of all collections and all heritage sites. In my preparation for this proposal, I did my research, and two things emerged which attracted my attention.

First, that this is not merely a business database which has an expected lifetime of around seven years before it is replaced with the old data often discarded. This National Inventory must live forever, and living forever is surprisingly difficult in the digital world. It was from this experience that I considered the vision of an eternal heritage, and I used the letters in this simple term to create the mnemonic for our business unit – ETHER : ETernal HERitage.

Secondly, that museums contain all the worlds cultural and natural history, covering a vast array of subjects, each with their own standards, and that to be successful, this National

Inventory must be able to handle all these subjects and how they structure their knowledge.

My area of expertise in AI is Knowledge Representation, essentially the dual questions of how we humans stored and retrieve our knowledge, and how we should build AI systems to replicate these processes. Well – the current AI systems certainly do the second of these, but we have remarkably failed completely on the first of these.

It was from this analysis that I prepared my proposal, submitted this, and won the tender, with a large team preparing for a new national inventory, and data capturing of every collection in the country. Sadly today, 21 years later, only a small fraction of these collections has been inventoried and brought into this aggregator. But the dream must remain, and rather than just one country, this should be an aggregator of all museum collections throughout the world. My approach was based on a Federated Model, in which minimal data was shared, and the rest retained by the organisations, like Europeana. For this I designed a data interface standard for the exchange of information, and then a data coding standard to handle every item of data as required. I referenced CIDOC and others as reference models, while I addressed the practicalities of actually building a database.

So where does AI fit?

I see that AI can assist in mining the inventories of museums, and WITHOUT having to structure these inventories into some standard model. Rather let AI do its job in sorting this out for us, supporting a search across all museums of the world without us having to do anything but build our inventories in the way we choose to do. No preferred systems, no interfaces needed, and to then arrive at the ability to query this entire world inventory as though it is One Museum.

I note at this point my own initiative to build this has been called the One Museum for the past 10 years, and is intended as a free-for-all aggregator of content or all types forms, in which users see the world of museums as though it is a single entity to be examined, while ensuring that all museums retain their complete independence the details of their collections.

For me, the problem is not in building this, since that I have already done, the problem is in populating the databases and the repositories, and this is where AI can play a significant role. AI can assist in taking an inventory and catalogue, in any form, and preparing the inputs into the One Museum, or to any other aggregator. But this requires one specific problem which is much larger...

Addressing Terminology

The Spectrum standard refers to many vocabularies which can be used to support collections management practices. However, it assumes that each museum will build their own vocabularies, which has the risk that there will be different terms used for the same concepts in every museum, even on Spectrum-compliant systems.

Collections Trust started an initiative for common vocabularies some years ago, but this is a massive undertaking and I am not sure where this went, but I hope that it is still ongoing.

I was commissioned to develop the National Terminology Policy for South Africa in 2024, and I explored this to the full, including the role of ISO 704 on Terminology Work, which forms a baseline for domain-specific terminologies.

With a common terminology including vocabularies, taxonomies, and nomenclatures, we have the basis for all speaking the same language, but this is a decades-long exercise in knowledge representation and knowledge management, and we must talk to each other now if we are to build an aggregator of all museum collections.

To the best of my knowledge, this problem remains very far from a practical solution, and so we need help.

Enter AI.

Whereas this problem is beyond AI as it currently stands, it can do the second-best thing (over the first best thing of all using the same terminologies) which is to interlink different terminologies under the same search and query processes. And this is a problem which is highly suited to AI.

Selling the idea that Museums are Trusted Sources

If these prior problems can be handled using AI, then I would like to add one more, which is counter to AI in its generative models – to sell the basic fact that museums can be trusted, when almost everything else in the world cannot, including the AI systems themselves.

Perhaps the largest mantra I have repeated in my presentations to museums and at conferences is for the museum sector to gain the high ground which you already own. You can be trusted, much in the same way as archives, reference libraries, scientific organisations, and research organisations. You have the unique attribute of trust which is core to your enterprise. You tell the truth and only the truth, and yet this small aspect of being a museum is often forgotten.

So what can AI do here. Absolutely nothing – and let us leave it that way.

Meeting our Generational Responsibilities

You most likely work in a museum if you are reading this. You may work there for a few years or for your entire working life. But no matter how long you work at the museum, you are always only a temporary custodian of the collection, which is the heart of the museum, and you may make improvements in building the collection, in preserving the collection and in making the collection accessible. But eventually you must hand over this responsibility to the next generation.

And this is where it often falls down, since in every case I have come across to date, the wealth of institutional memory held by the experts who live the museum over long time periods were never mined to assist the next generation. Even when reports are written by these experts to document parts of the museum, these are often ignored by the newly appointed curators, who want to sweep clean. So much is lost.

So how can AI assist in accessing this information?

My suggestion is for the experts to record what they do, in audio or video, and to then pass this onto an AI with the purpose of documenting their legacy. Like the work done in some museums to provide holograms of people who have passed, such as in the holocaust, who tell their stories from the grave in response to questions from the visitors. This is no longer high tech, and anyone can generate videos for this, but these lack the authenticity of the true voice.

So, get those audio recorders out, and say anything and everything into this, no matter how small, such as a small story about an object. You do not need to be concerned about whether this will be read – it will, and it will be read by an AI, which will then use this knowledge base as a searchable source directly from the voice of the previous curator. Perhaps the outgoing curator may not like their knowledge being reused, and this opens the question of who owns this knowledge, but my money is on the museum owning everything, and I expect that every retired museum curator would rather have their contribution acknowledged rather than the opportunity to sell themselves back to the institution to give back what was never recorded at the time they left.

The End – and a New Beginning

Whereas this is the end of my little story, this is also the start of the work itself. I never like to write things which go nowhere. For me there is always a next step, and this is mostly phrased as a Call-To-Action.

AI will play a major part in our future, and my personal use is high, a few hours per day, but I use AI as a tool and nothing else. And using this tool I can do things which would have taken me one hundred times longer if I were to do this without this help. I can now get to all those projects I have put aside and take giant steps in a few hours.

So let us find a way to work together, with the help from the various AIs of the world, to meet our generational responsibility – which is little other than to prepare the way for the those who will take our place.

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